Indian Institute of Technology, Kanpur
Proposal for a new course

Course Number: ME 6XX

Course Title: Introduction to Tribology

Credits: 3L-0T-0P-0A: 9

Course level: Open elective (PG and 3rd & 4th year UG)

Duration of Course: Full Semester

Proposing Department: Mechanical Engineering

Other departments which may be interested in the proposed course: Materials Engineering, Chemical Engineering, Aerospace Engineering

Proposing Instructor: Manjesh K. Singh

Other faculty who might be interested in teaching this course: Dr. Tanmay Mathur, Dr. Ushasi Roy, Dr. Sudhansu Shekhar Singh, Dr. J. Ramkumar, Dr. Kantesh Balani, Dr. Ishan Sharma

Course Description

(a) **Objective:** To introduce the students to fundamentals and different components of tribology− friction, wear and lubrication.

(b) **Contents** (assuming 50 minutes lectures, total 40 lectures):

(i) **Introduction** [1 Lecture]
Definition, history and economic cost

(ii) **Friction and Adhesion** [12 Lectures]

- Surfaces: Roughness, real and apparent contact
- Adhesion: Surface-forces, bonds and free surface energy theory of adhesion
- Contact of spheres: The Hertz theory, the JKR theory and the DMT theory
- Relative motion of surfaces: Sliding, rolling and spin
- Origin and laws of friction
- Friction in sliding/rolling contact

(iii) **Lubricants and Lubrication** [10 Lectures]

- Properties of lubricants: Viscosity and thermal properties
- Fluid film lubrication: Viscous flow and Reynolds equation, Stribeck curve, boundary, mixed and hydrodynamic regime
- Hydrostatic, hydrodynamic and elastohydrodynamic lubrication
- Boundary lubrication and lubricants
- Solid lubrication and lubricants

(iv) **Wear** [8 Lectures]

- Wear and its causes
- Adhesive, abrasive, erosive, fatigue, corrosive and fretting wear
- Mathematical modelling of wear

(v) **Special topics**: Instructor's discretion to focus on some of the following topics. [9 Lectures]

- Friction and wear measurements: Laboratory experiments/demonstrations.
- Design of bearings and gears
- Tribology in Automobiles, Aircrafts and Electric Vehicales
- Biotribology: Natural and artificial hip joints
- The JKR-DMT transition
- Computational Tribology

(c) **Prerequisites:** None

Books/References

1. Israelachvili, J. "Intermolecular and Surface Forces", 3rd Ed., Academic Press
2. Bhushan, B. "Introduction to Tribology", 2nd Ed., Wiley
3. Stachowiak, G. W. and Batchelor, A. W. "Engineering Tribology", 3rd Ed., Butterworth-Heinemann
4. Hutchings, I. and Shipway, P., "Tribology", 2nd Ed., Butterworth-Heinemann
5. Johnson, K. L., "Contact Mechanics" , Cambridge University Press
6. Goryacheva, I. G., "Contact Mechanics in Tribology", Kluwer Academic Publishers
7. Maugis, D., "Contact, Adhesion and Rupture of Elastic Solids", Springer

Dated: 19-10-2024 Proposer: Manjesh K. Singh

Dated: _____ DUGC/DPGC Convener: _____

The course is approved/ not approved

Chairman, SUGC/SPGC **Dated:** _____